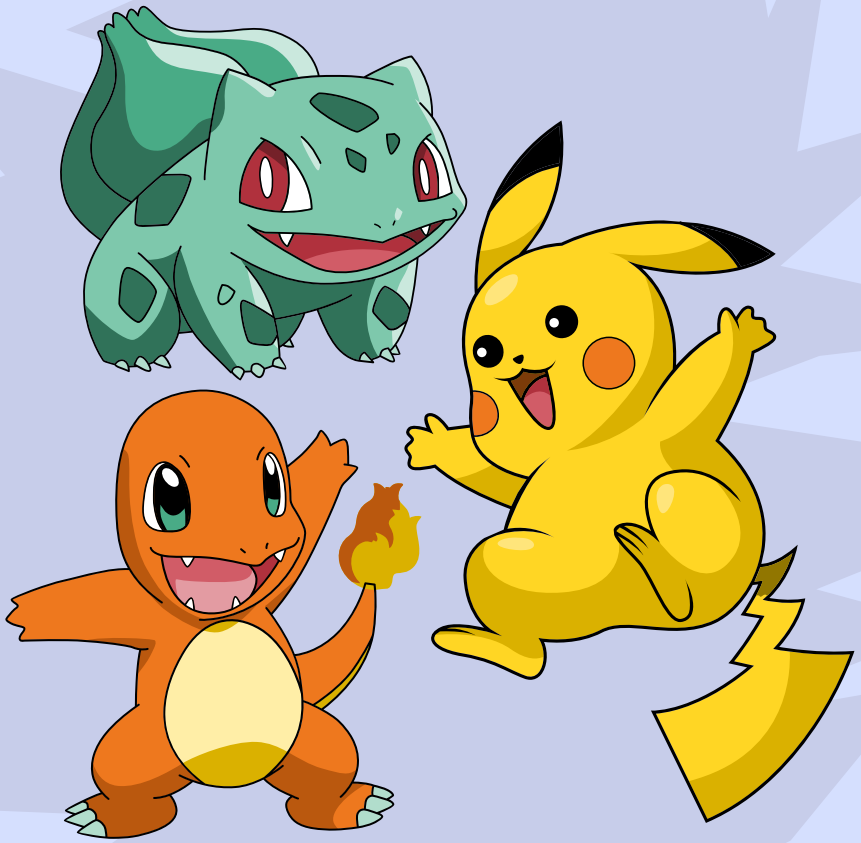


POKÉMON

DATA ANALYSIS

Meet The Optimizers

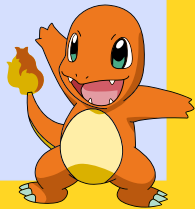
Us with our clients

Research Question

By analyzing survivability across various Pokémon, how do trade-offs emerge between stats, and what patterns emerge across generations, types, and legendary status?

Executive Summary

- Dataset: 1,025 Pokémon analyzed.
- Survivability Index = HP + Defense + Special Defense.
- Legendary Pokémon generally have higher survivability.
- Certain type combinations result in stronger survivability.
- Survivability does not negatively correlate with offensive stats.
- Evidence of power creep across generations.



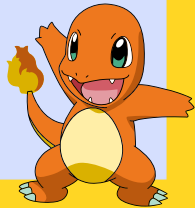
Goal

To quantify survivability and explore patterns by type, generation, and legendary status.

Make sure our clients are happy and successful.

Primary Metrics

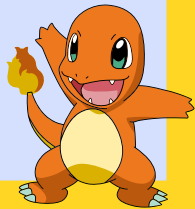
Survivability Index = HP + Defense + Special Defense.



Setup & Data Hygiene

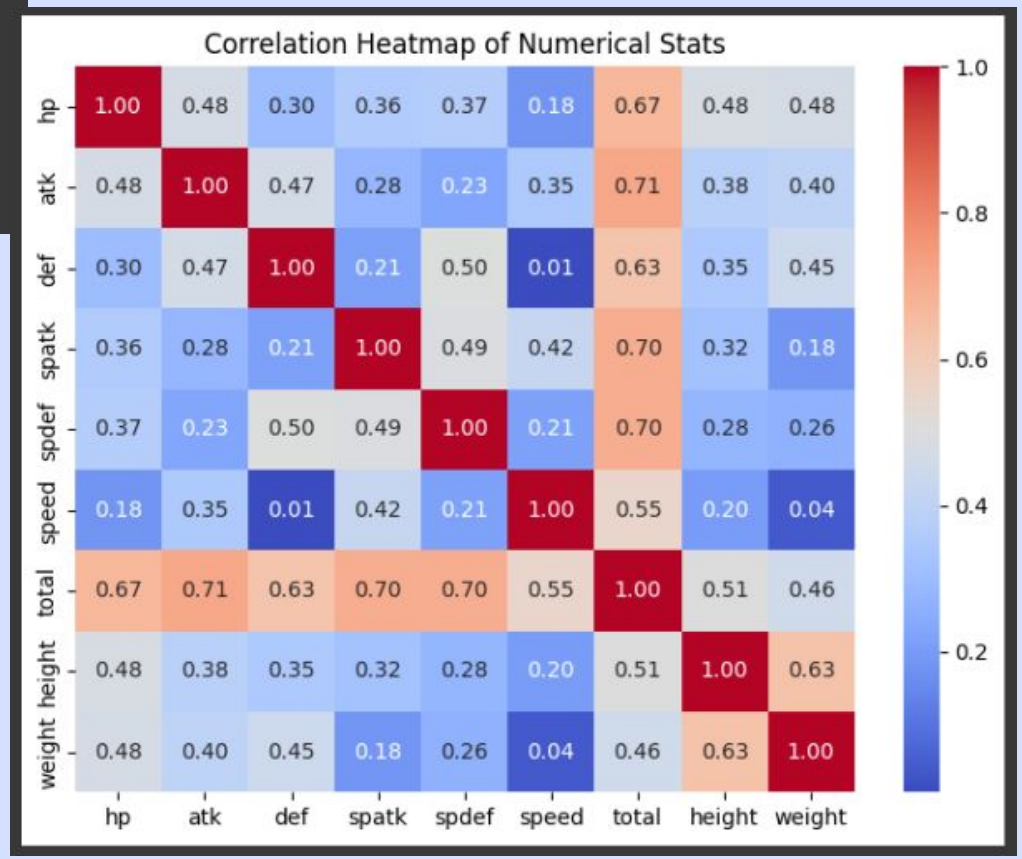
```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

url = '/content/pokemons.csv'
pk_df = pd.read_csv(url)

pk_df.head()
pk_df = pk_df.drop_duplicates()
pk_df = pk_df.fillna('')

# add a column based on whether or not the rank is legendary
pk_df['is_legendary'] = pk_df['rank'] == 'legendary'
```

- Libraries imported: NumPy, Pandas, Matplotlib, Seaborn for analysis and visualization.
- Dataset loaded: 1,025 Pokémon stats imported from CSV file.
- Cleaning performed:
  - Removed duplicate entries.
  - Filled in missing values.
- Feature engineering: Added is_legendary column to distinguish Legendary Pokémon.
- Ensures dataset is reliable, consistent, and ready for survivability analysis.



Correlation between Pokemon

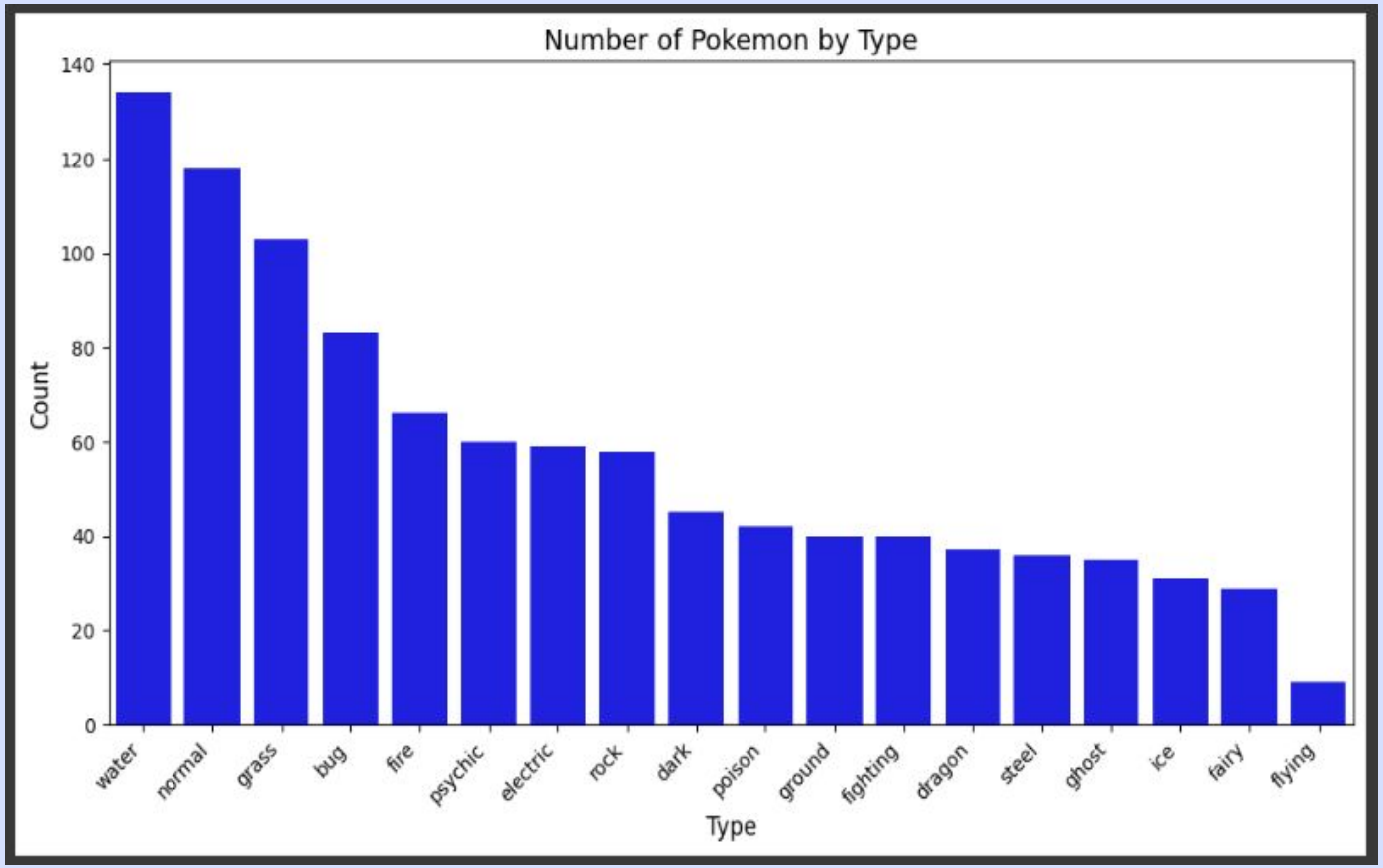
```
# heatmap for each numerical stat in the dataframe
corr = pk_df[cols].corr()

plt.figure(figsize=(8,6))
sns.heatmap(corr, annot=True, cmap="coolwarm", fmt=".2f", cbar=True)
plt.title("Correlation Heatmap of Numerical Stats")
plt.show()
```

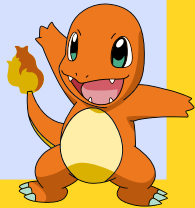


- No negative correlations between stats.
- Increases in one stat often accompany increases in others.
- Suggests Pokémon design balances stats upward rather than forcing strict trade-offs.

Number of Pokemon by Type

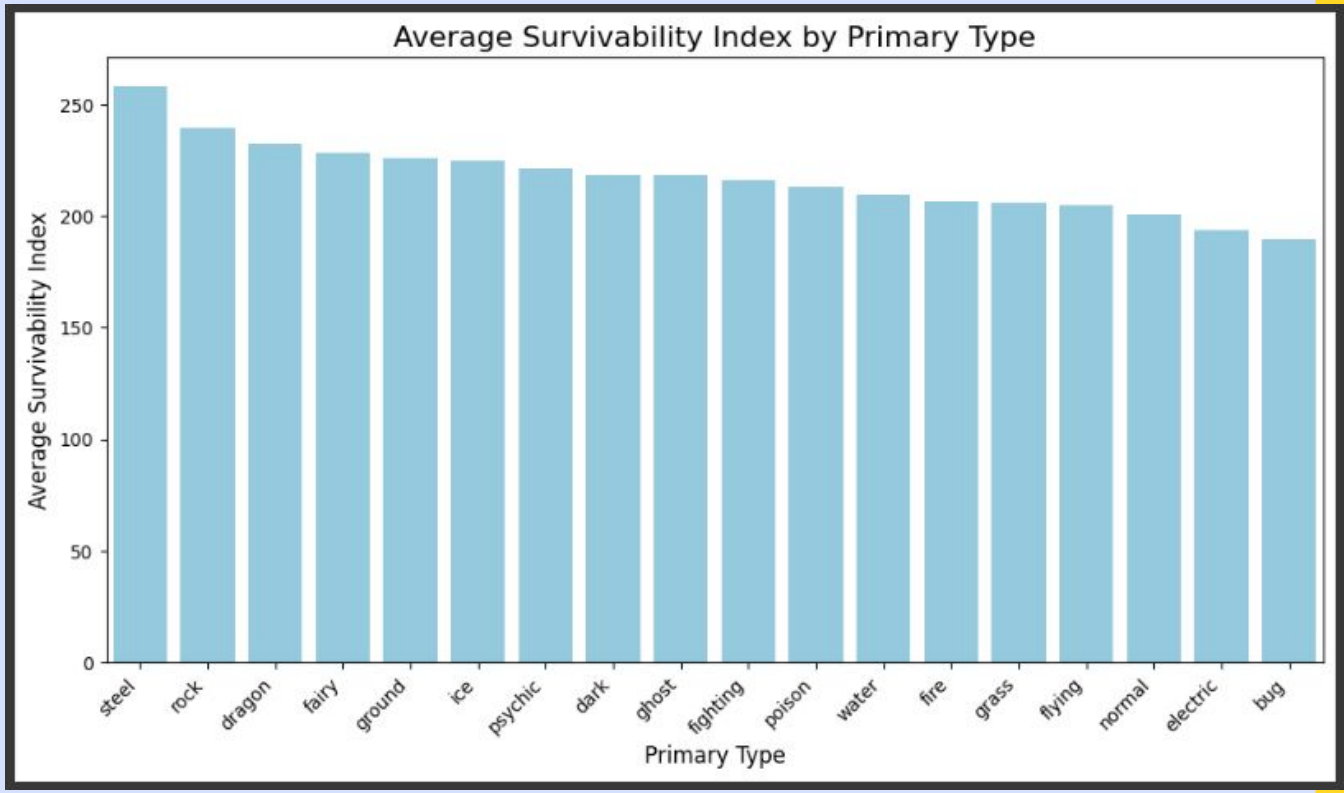
- Water is the most common type, with over 130 Pokémon.
- Normal and Grass types follow as the next most frequent.
- Middle-tier types include Bug, Fire, and Psychic.
- Least common types are Ice, Fairy, and Flying (as primary types).

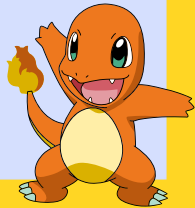




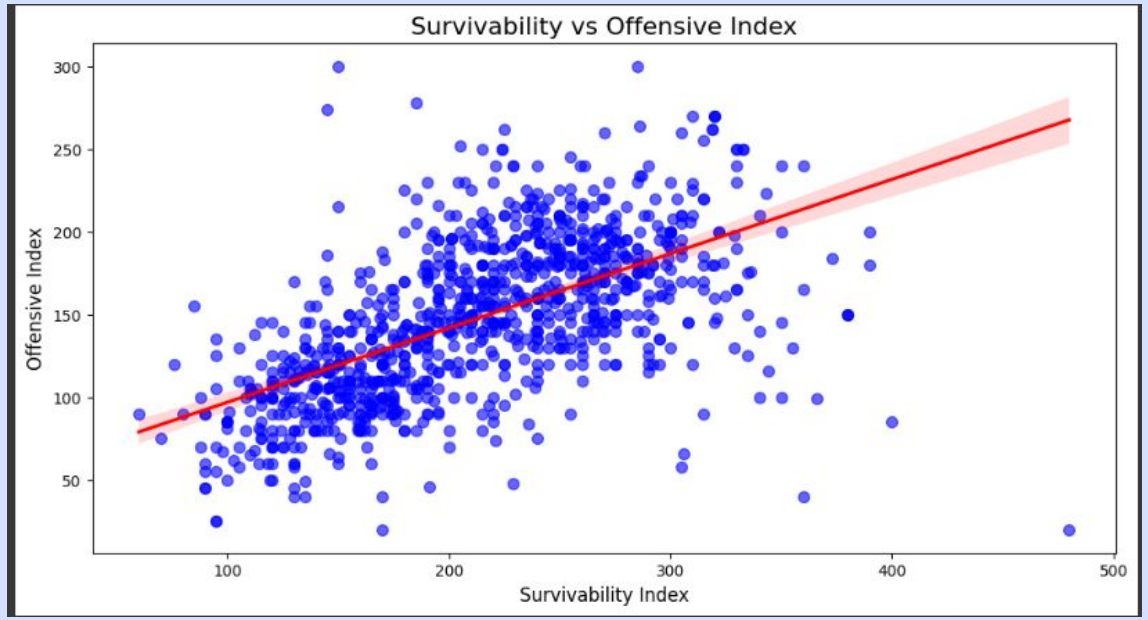
Average Survivability Index by Primary Type

- **Steel** Pokémon had the highest survivability.
- **Bug** Pokémon had the lowest survivability.
- **Rock, Dragon, and Ground** types also ranked high, matching their durable reputation.
- Results show that **type identity heavily shapes survivability design**.



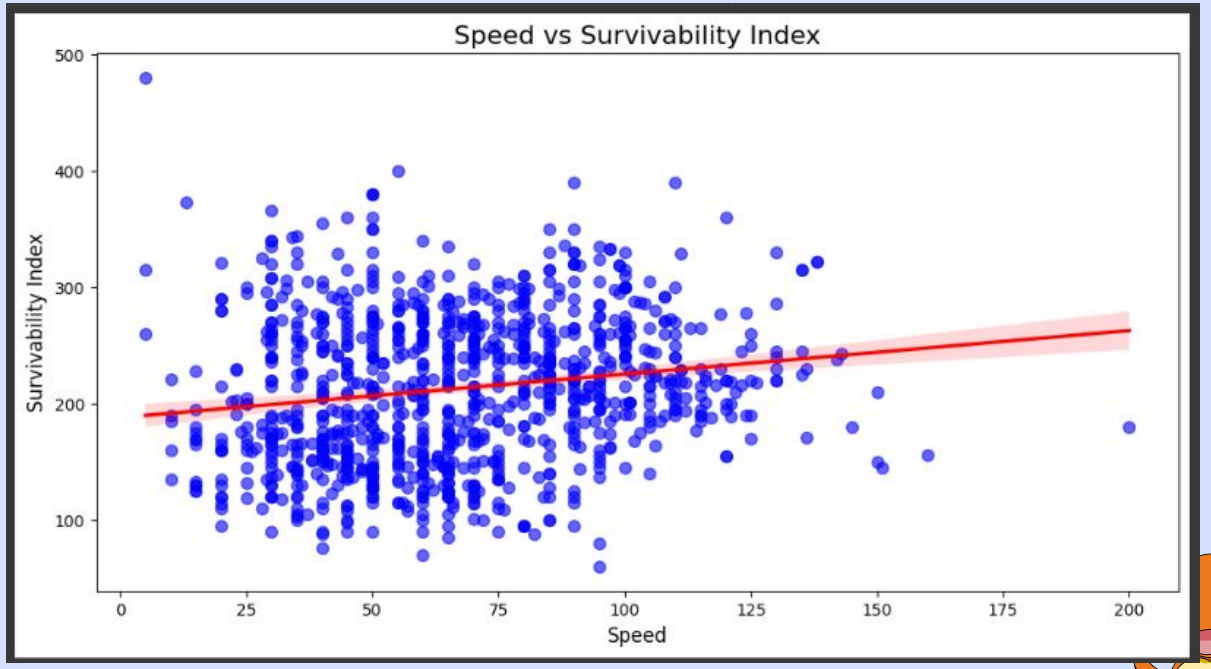


Survivability Visualizations



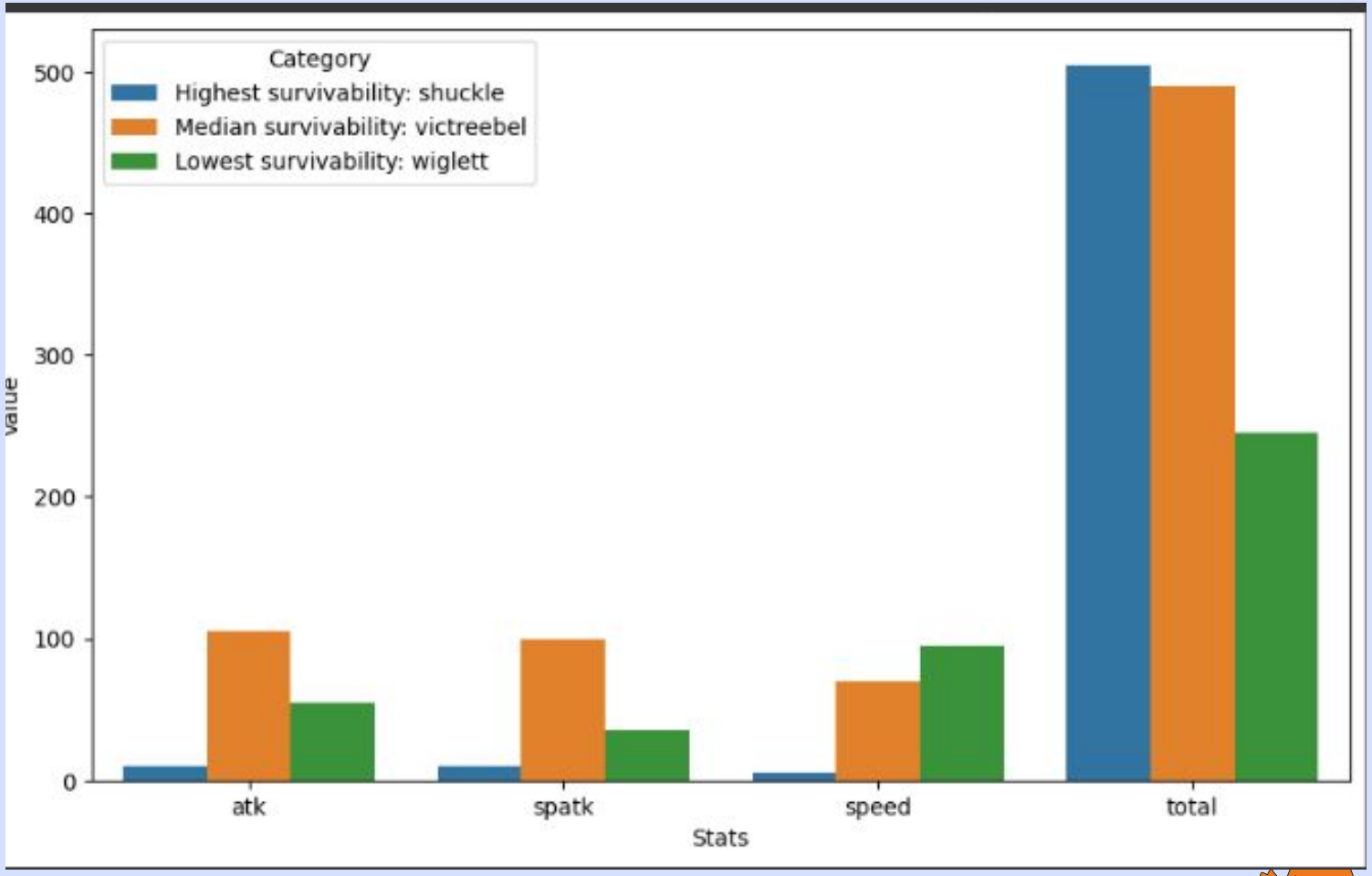
- No negative correlations between stats.
- Increases in one stat often accompany increases in others.
- Suggests Pokémon design balances stats upward rather than forcing strict trade-offs.

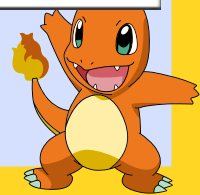
- High survivability does not limit the speed of a Pokemon despite the intuition that high defense may limit movement.
- Instead as survivability increases so does speed, though at a low rate



High vs Median vs Low

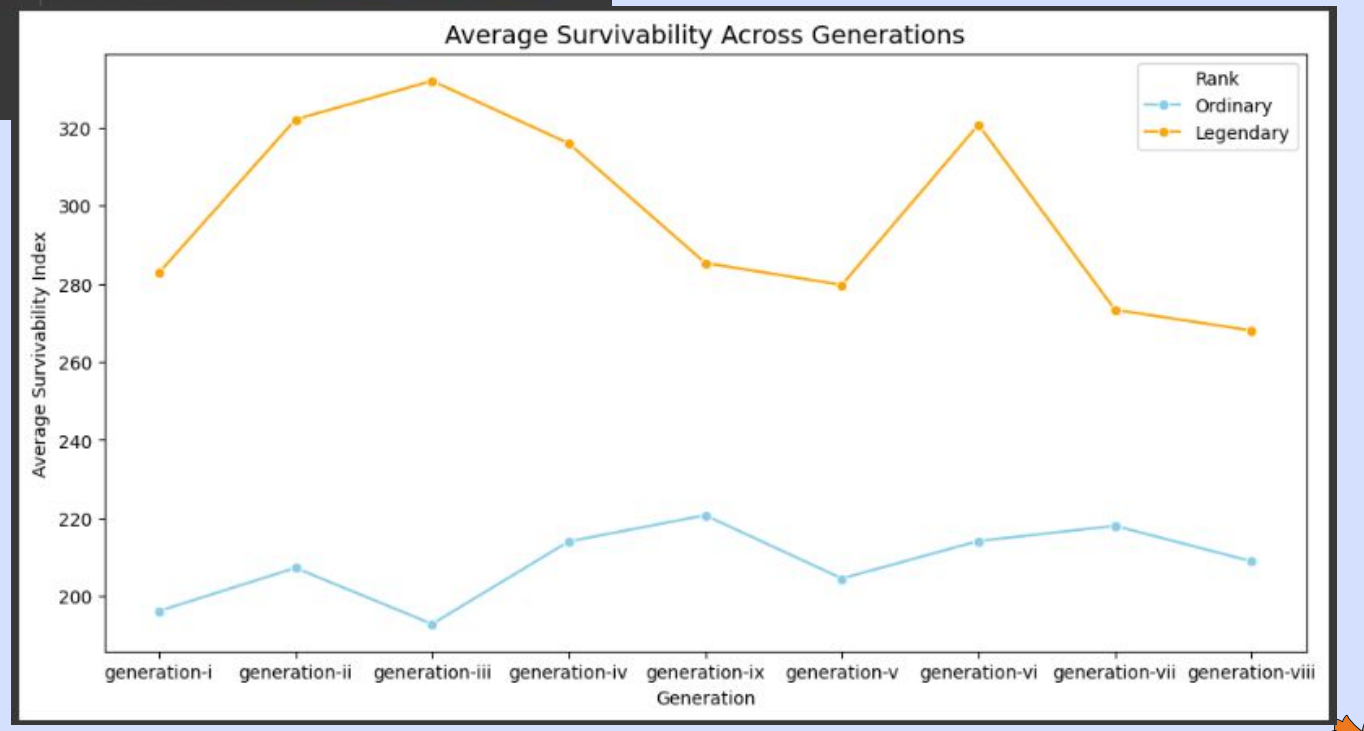
- **Shuckle** has the highest survivability index.
- **Victreebel** represents the median survivability.
- **Wiglett** has the lowest survivability index.
- Survivability levels don't directly predict total stat balance (e.g., Wiglett's speed offsets its low defense).



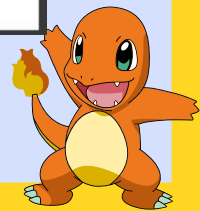


Average Survivability Across Generations

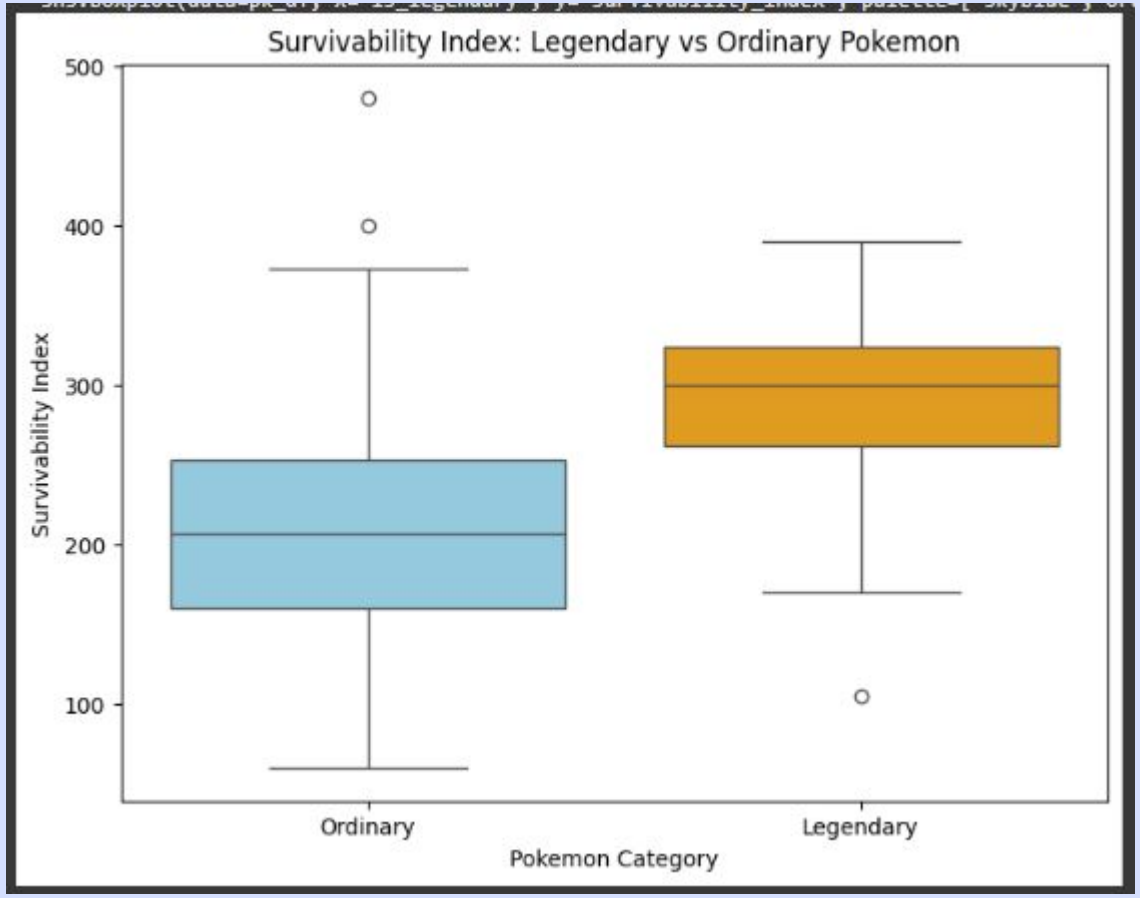
```
# this graph focused on visualizing potential power creep in the survivability index
# in other words, whether or not survivability increased as the generations went on
avg_surv = pk_df.groupby(['generation', 'is_legendary'])['survivability_index'].mean().reset_index()

avg_surv['Rank'] = avg_surv['is_legendary'].replace({False:'Ordinary', True:'Legendary'})
# in the line plot ordinary and legendary pokemons were differentiated to see how stat inflation affected each individually
plt.figure(figsize=(12,6))
sns.lineplot(data=avg_surv, x='generation', y='survivability_index', hue='Rank', marker='o', palette=['skyblue','orange'])
plt.title("Average Survivability Across Generations", fontsize=14)
plt.xlabel("Generation")
plt.ylabel("Average Survivability Index")
plt.show()
```



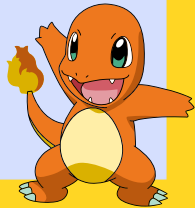
- Legendaries peaked in survivability in Generation III.
- Non-legendaries peaked in Generation IX.
- Survivability fluctuates → Pokémon design evolves over time.



Legendary vs Ordinary Pokemon

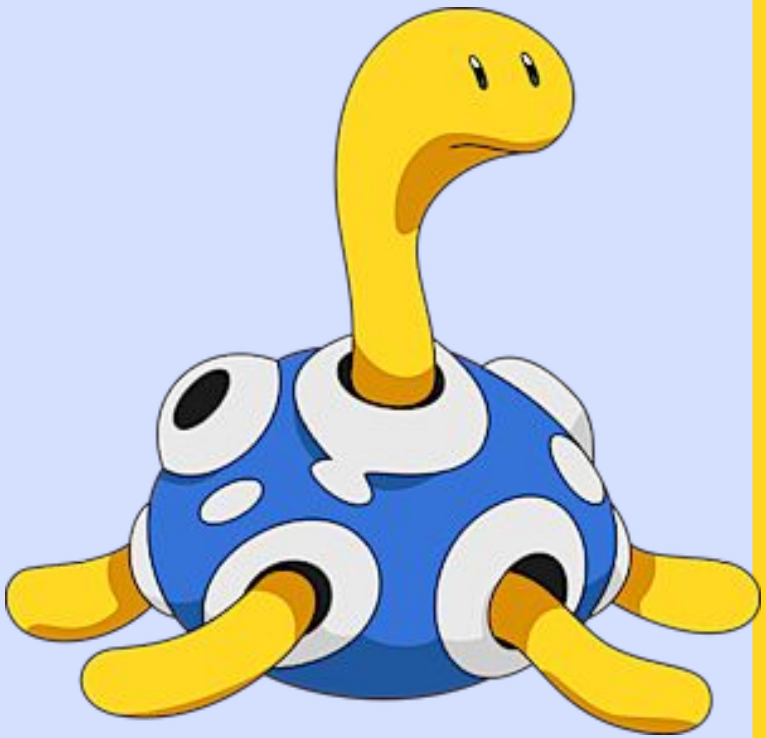


- Legendaries generally score higher survivability.
- Some ordinary Pokémon outmatch weaker legendaries.
- Legendary status ≠ guaranteed tankiness.



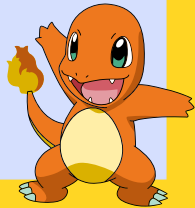
Outlier Analysis

- Shuckle highlights the tradeoffs between offense and survivability.
- Shuckle has a survivability index of 480 due to its high Sp. Defense and Defense.
- The Pokémon is limited by its low offense, with an Attack and Sp. Attack of 10.
- This suggests that high survivability does not guarantee capability in battle



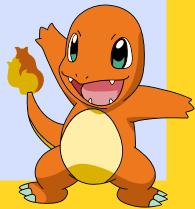
Key Insights

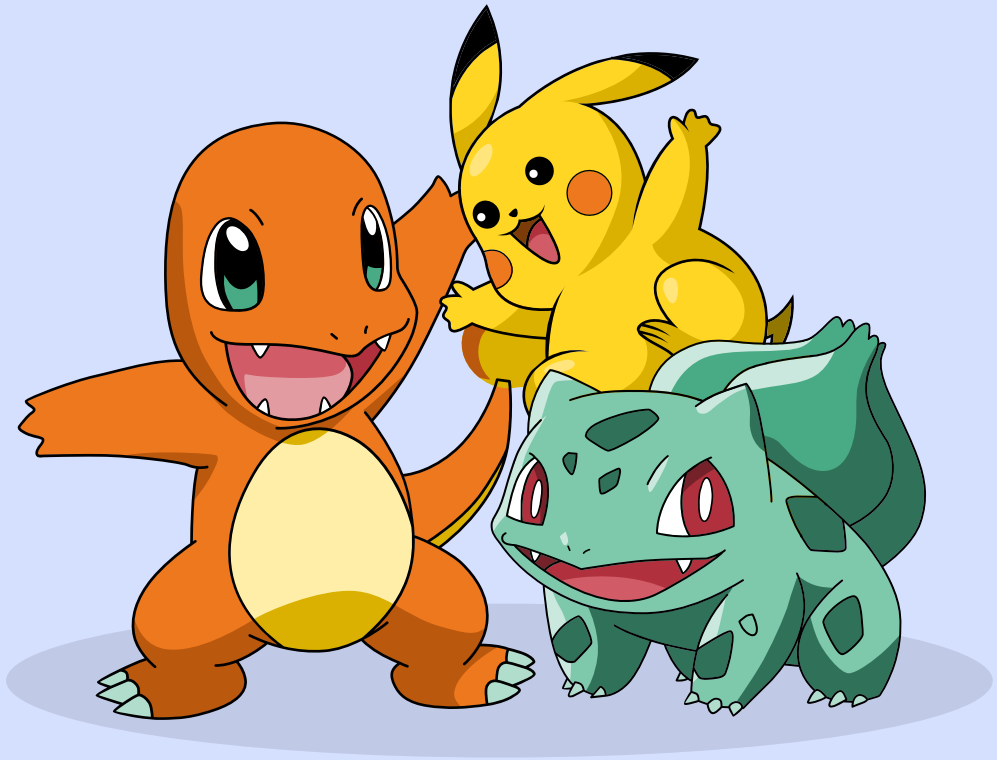
- Survivability Index (HP + Defense + Special Defense) reveals clear trends across Pokémon.
- Legendary Pokémon generally have higher survivability, but strong non-legendaries exist.
- “Power creep” is evident, survivability peaks shift across generations, showing evolving design.



Limitations and Future

- Develop a **weighted survivability index**, adjusting HP, Defense, and Sp. Defense for context.
- Run **battle simulations** to evaluate how survivability plays out in practice.
- Study **synergy with offensive stats** and explore survivability in competitive formats.
- Extend research to **Mega Evolutions, regional forms, and other Pokemon** to see if mechanics shift survivability patterns.





THANKS!

for Joining Our Pokémon Survivability Investigation!

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